



IF YOU LIKE BRAIN TEASERS, THEN YOU'RE IN FOR SOME FUN!
THE OBJECT OF THIS PUZZLE IS TO END UP WITH A 3x3 MATRIX THAT HAS 1's
IN ALL POSITIONS EXCEPT THE CENTER WHICH WILL BE 0.

THE POSITIONS ON THE BOARD ARE REFERRED TO AS ROW, COLUMN.
FOR EXAMPLE, THE UPPER RIGHT HAND CORNER IS 1, 3.

HERE ARE THE RULES:

YOU MAY CHANGE ANY 1 TO A 0. YOU'RE NOT ALLOWED TO CHANGE
A 0 TO A 1.

WHEN YOU MOVE TO CHANGE A 1, THERE ARE "SIDE" EFFECTS.

MOVING IN A CORNER

DRAW A 2x2 BOX CONTAINING THE CORNER. THEN EACH
POSITION IN THE BOX CHANGES STATE.

MOVING IN THE CENTER OF AN EDGE

EACH POSITION ALONG THAT EDGE CHANGES STATE.

MOVING IN THE CENTER POSITION.

EACH POSITION IN A '+' (PLUS SIGN) CHANGES STATE.

TRY THE GAME A FEW TIMES AND SEE WHAT HAPPENS ON THE BOARD.
THE NEW BOARD WILL BE PRINTED AFTER EACH MOVE.

HINT (CHUCKLE): IT IS POSSIBLE TO REACH THE GOAL FROM EVERY POSITION
(EXCEPT ALL 0's) WITHIN 11 MOVES, IF YOU CAN FIND THE RIGHT MOVES.

HERE WE GO ... THE BOARD IS

```
0 1 0
1 0 1
0 0 0
```

YOUR MOVE? 2, 3

```
0 1 1
1 0 0
0 0 1
```

YOUR MOVE? 3, 3

```
0 1 1
1 1 1
0 1 0
```

YOUR MOVE? 2, 1

```
1 1 1
0 1 1
1 1 0
```

YOUR MOVE? 2, 2

```
1 0 1
1 0 0
1 0 0
```

YOUR MOVE? 3, 1

```
1 0 1
0 1 0
0 1 0
```

YOUR MOVE? 3, 2

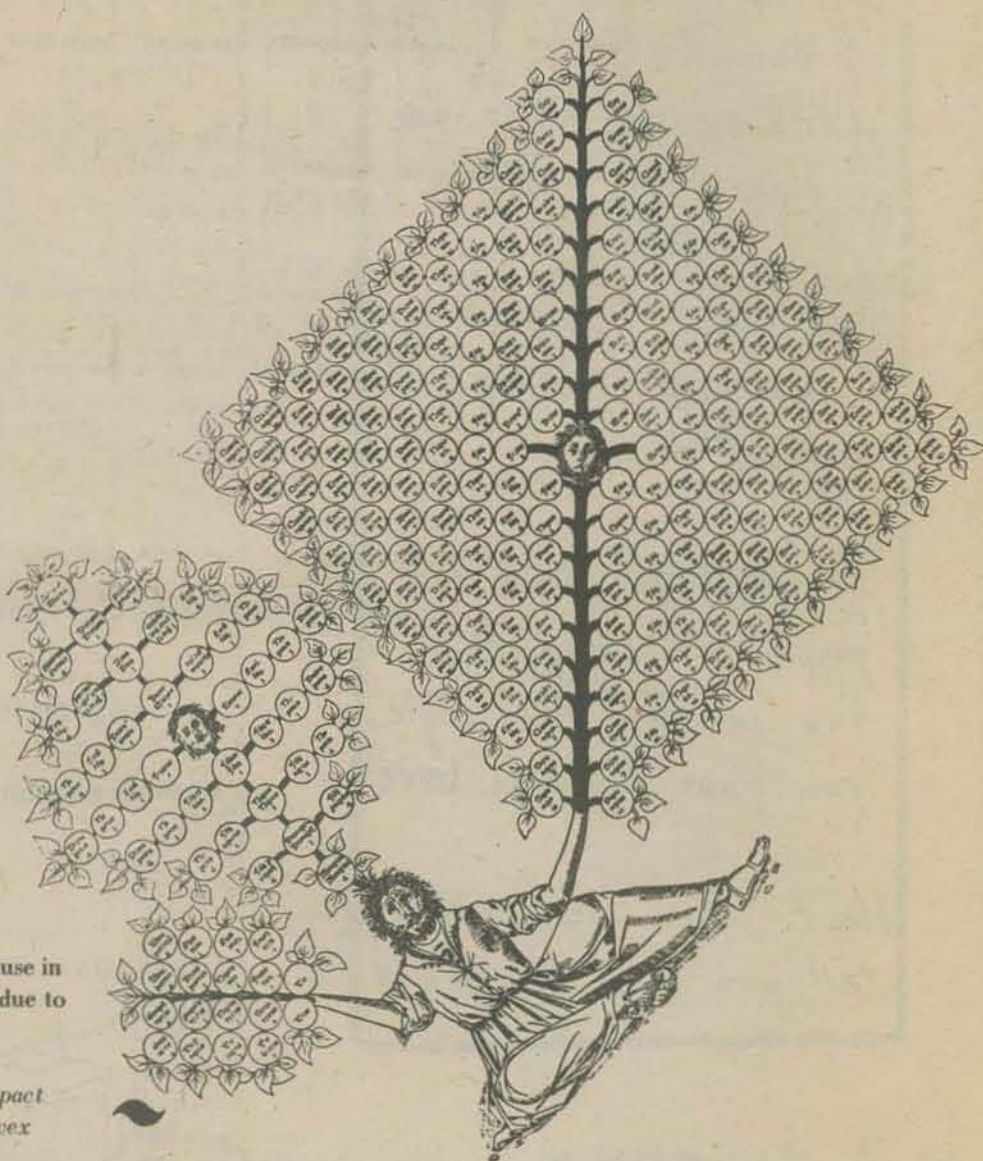
```
1 0 1
0 1 0
1 0 1
```

YOUR MOVE? 2, 2

```
1 1 1
1 0 1
1 1 1
```

One theorem we did NOT use in
analyzing TEASER is one due to
Krein and Milman:

Theorem: Let K be a compact
convex set in a locally convex
topological vector space X .
Then K is the closed convex
hull of its extreme points.



A COMPLETE ANALYSIS = of = TEASER

THIS DIAGRAM CONTAINS
EACH OF THE 102 POSSIBLE
POSITIONS (EXCLUDING
ROTATIONS & REFLECTIONS).

